POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

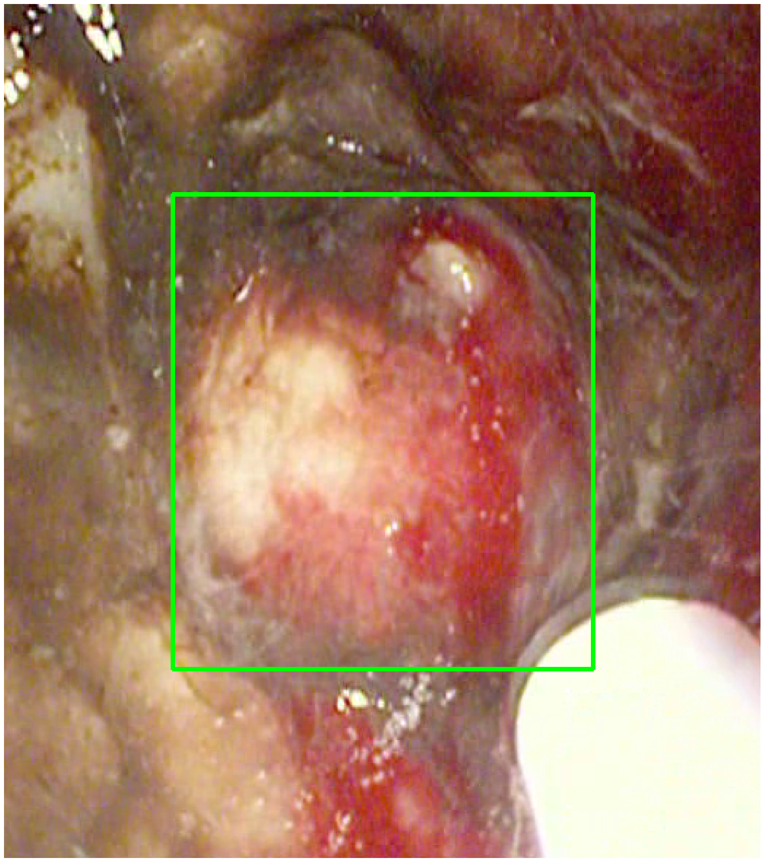
Video ID:	2eb34265-6359-4233-bdbb-71f3389e220a
Total Frames:	3006
Frame Rate:	25.0 fps (assumed)
Duration:	120.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	96	✓
RT-DETR Detections	941	✓
MedSAM2 Detections	942	✓
Consensus (3-Model Agreement)	5	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	5	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 315 frames)



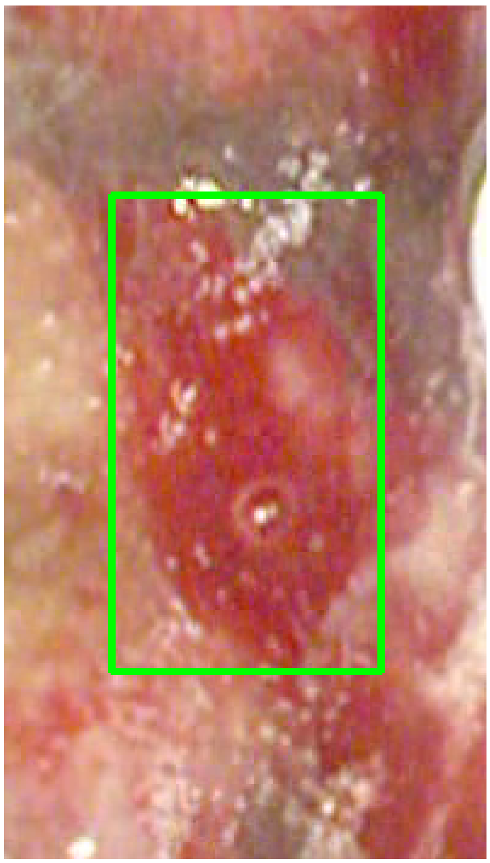
Feature	Value	Interpretation
Frame Index	1042	Frame 1042 of 3006
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 90.9%
Redness Score	0.159	Color-based vascularization
Relative Radius	16.9%	Polyp size as % of frame diagonal
Texture Score	0.038	Surface morphology
Vessel Visibility	0.504	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	65.0%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 315 frames)



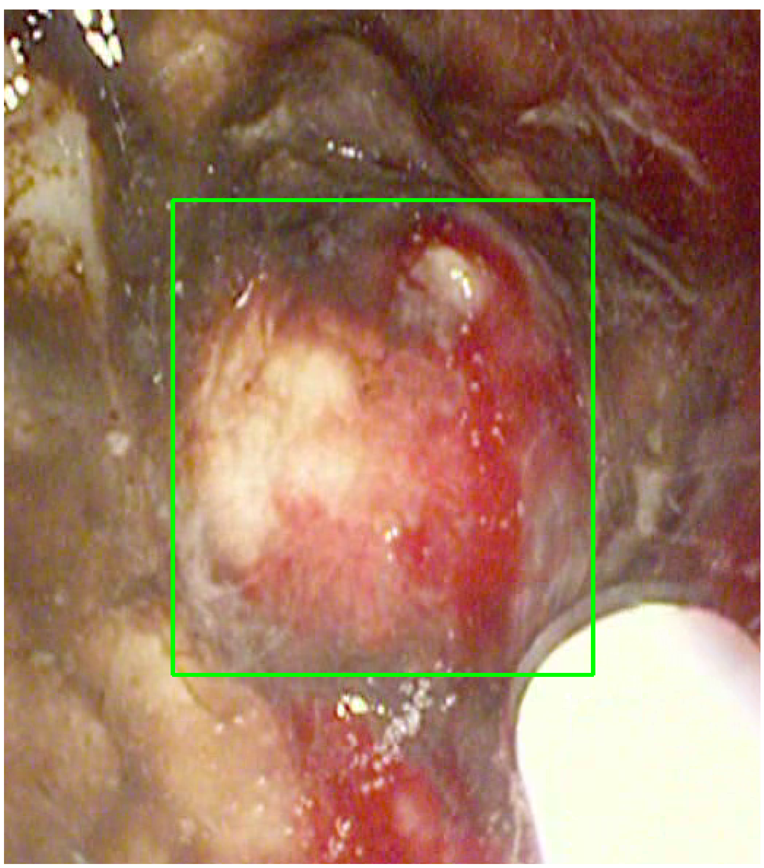
Feature	Value	Interpretation
Frame Index	1042	Frame 1042 of 3006
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**FLAT_POLYP**	Type confidence: 90.9%
Redness Score	0.159	Color-based vascularization

Relative Radius	16.9%	Polyp size as % of frame diagonal
Texture Score	0.038	Surface morphology
Vessel Visibility	0.504	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	59.9%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 148 frames)



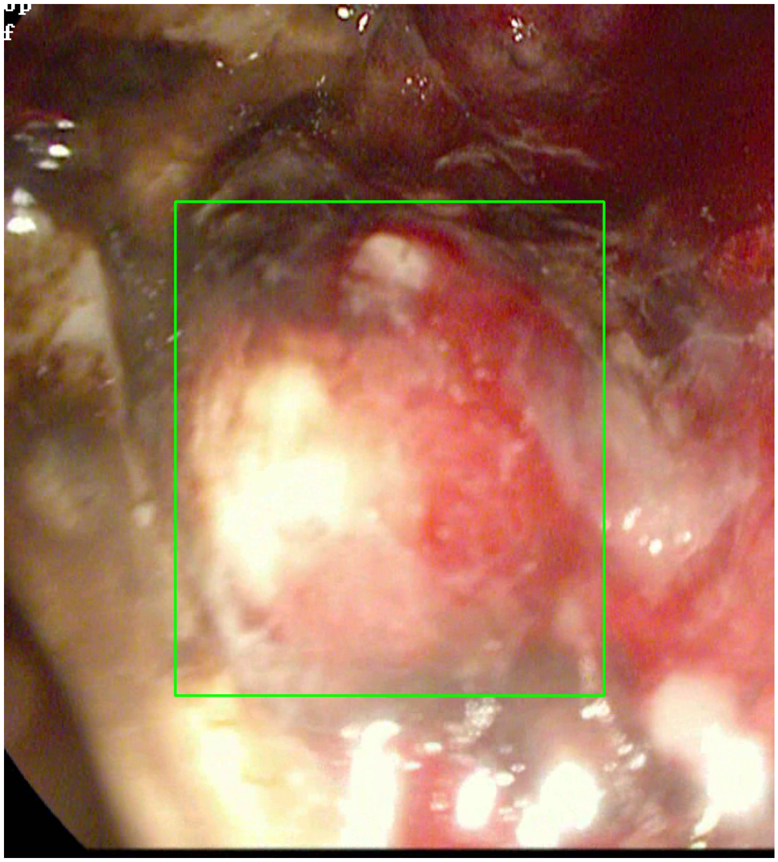
Feature	Value	Interpretation

Frame Index	1042	Frame 1042 of 3006
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.281	Color-based vascularization
Relative Radius	11.2%	Polyp size as % of frame diagonal
Texture Score	0.055	Surface morphology
Vessel Visibility	0.953	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	61.8%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #4 — SMALL_POLYP [LOW_RISK] (Tracked across 34 frames)



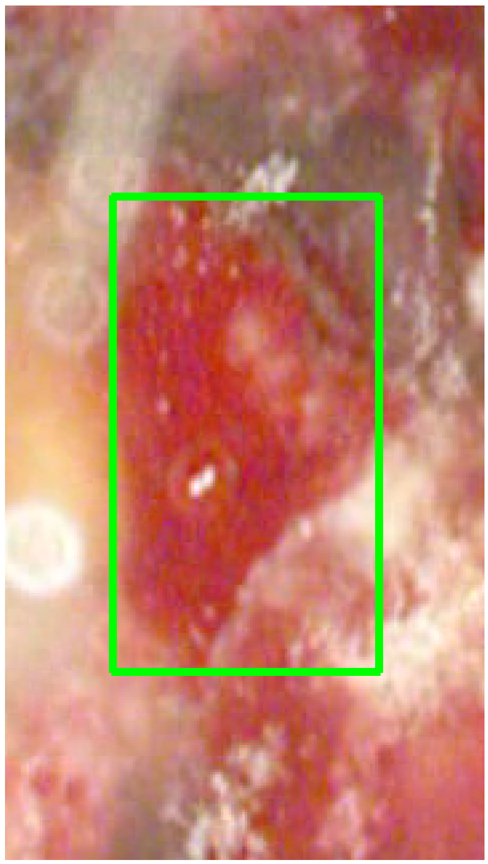
Feature	Value	Interpretation
Frame Index	1420	Frame 1420 of 3006
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**SMALL_POLYP**	Type confidence: 92.0%
Redness Score	0.232	Color-based vascularization
Relative Radius	6.6%	Polyp size as % of frame diagonal
Texture Score	0.023	Surface morphology
Vessel Visibility	0.389	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	56.8%	YOLO / RT-DETR / track confidence

Type Assessment: Diminutive lesion — resect and discard per protocol.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #5 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 111 frames)



Feature	Value	Interpretation
Frame Index	1288	Frame 1288 of 3006
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.356	Color-based vascularization

Relative Radius	10.2%	Polyp size as % of frame diagonal
Texture Score	0.069	Surface morphology
Vessel Visibility	0.958	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	61.4%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:
A total of 5 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

AI-Based Polyp Detection System
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